CS 6210 Spring 2026 Test 3

(120 min Canvas Quiz)

[Total Points: 74]

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Internet Computing [16 points]

Giant Scale Services [4 points]

1. [4 Points] You are an engineer working on a search service that can sustain 60,000 shard visits per second. Under normal operation, each query visits 120 shards, so the service sustains 500 queries per second.

Assume the traffic spikes to 750 queries per second, and you want to preserve yield.

a) [2 Points] What is the maximum data/work per query, Dv, your system can afford during the spike? Briefly show your work.

Ans: [+2] Dv = 60,000 shards per sec / 750 queries per sec = **80** shards per query

b) [2 Points] If each shard contributes equally to answer completeness, what is the harvest during the spike? Briefly show your work.

Ans: [+2] Reduced shards used per query / full shards per query

= 80 / 120 = 2/3 or 0.667

Map Reduce [6 points]

1. [2 points] (Answer True/False with Justification. No credit without justification)

Increasing the number of reduce workers more than the number of final desired outputs improves parallelism and therefore improves runtime.

Ans:

- False
- Reduce function is sequential in nature so adds no benefit to the runtime

2. [2 points] The master keeps a scoreboard of workers assigned to map tasks with the intermediate results of the mappers stored at the respective mapper nodes. A new engineer proposes centralizing all intermediate results at the master "to simplify reducer fetches."

Give ONE pro and ONE con of this proposal.

Ans:

- [+1] Any good reason: simpler design, or reducers fetch from one place, or easier bookkeeping.
- [+1] Any bad reason: master becomes a network bottleneck, or master becomes a storage/memory bottleneck, or master becomes a bigger single point of failure / scalability bottleneck.

3. [2 points] A novice user of the map-reduce framework, writes a Map function that inserts a row into an external database for every processed shard of input data before producing the intermediate result expected from each mapper.

With a few succinct bullets, explain why the user's application may not work correctly.

Ans: (any of the following bullets two receives full credit)

- A worker task assigned to a map function could crash after doing the DB inserts, but before producing the intermediate results. When the map task is re-assigned to a new worker there will be spurious duplicate entries in the DB.
- The Map function as written by the user breaks the idempotency requirement of the map-reduce framework

CDN [6 points]

1. [2 points] Giant-scale services introduced the terms "yield" and "harvest" through the DQ principle. A CDN that uses eventual consistency to keep the content of the mirror sites in sync is also an example of a giant-scale service. Explain with succinct bullets, how you can relate those terms to such a CDN?

Ans:

- Yield can be interpreted as the percentage of data retrieval requests that are serviced successfully by the CDN.
- Harvest can be interpreted as the percentage of data retrieval requests that return the up to date content from the CDN.

2 [2 points] Consider a CDN with DNS-based routing to route client queries to proximal mirror sites, DHT-based object lookup, and eventual consistency to keep the mirrored content in sync.

A news article is updated by the origin site from V1 to V2.

Answer True/False with justification. No credit without justification.

Two client queries may receive different versions of the news article

Ans:

- True
- DNS may route the queries to different mirror sites, which may have different versions due to eventual consistency.

3. [2 points] Answer True/False with justification. No credit without justification.

In the Coral DSHT, if a node is "full" for a specific key, it will reject all subsequent store requests for that key to protect its resources, effectively stopping any new references from reaching nodes that are even closer to the key in the ID space.

Ans:

- False

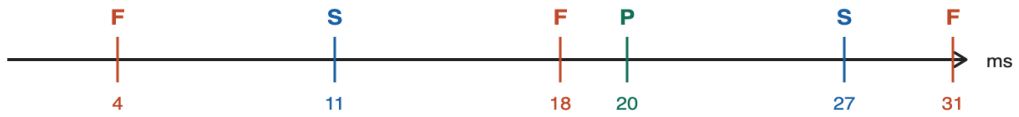
- The "leakage" principle: Even when a node is full/loaded, it allows a small fraction of requests to "leak" through to nodes closer to the key. This ensures the index can still be updated at the most optimal nodes (maintaining freshness) while the "sloppy" references handle the bulk of the flash-crowd volume.

Realtime & Multimedia [13 points]

TS Linux [5 points]

1. [3 points] Consider the timeline below.

P = periodic,
F = firm timer one-shot (overshoot = 3 ms),
S = system call



How is each F event handled by the OS given the above timeline?

Ans:

- [+1] F at t=4: no syscall/periodic within overshoot window before t=7. One-shot interrupt is fielded.
- [+1] F at t=18: periodic P at t=20 is within 3 ms overshoot window. Timer dispatched at t=20, one-shot interrupt avoided.
- [+1] F at t=31: syscall S at t=27 is before the window (28–34 ms). No soft-timer opportunity, one-shot interrupt is fielded.

2. [2 points] [True/False with justification. No credit without justification.] The lock-breaking preemptible kernel allows preemption even while shared kernel data structures are being modified.

Ans:

- [+1] False
- [+1] Preemption is inserted only after shared data is safely updated and the lock is released.

PTS [8 points]

1. [6 points] Jiahao is implementing a situation-aware application using PTS. There are three sensor sources: camera, audio from a microphone, and smoke detector. The camera images are accumulated in Channel Ch1; samples from the microphone are periodically recorded in Channel Ch2; smoke detector sample levels are recorded in Channel Ch3. Jiahao's application reads the most recent camera image; it then uses the timestamp associated with the camera image to get the closest (in time) samples from the other two channels.

Here are the states of the channels at some point of time:

- **Ch1** contains items with timestamps: 50, 100, 150, 200
- **Ch2** contains items with timestamps: 60, 120, 180
- **Ch3** contains items with timestamps: 30, 60, 90, 120, 150, 180, 210

Here are the APIs that PTS exposes to Jiahao:

- Get(ch, ts) returns the item with timestamp closest to ts
- Get(ch, "oldest") returns the earliest timestamp
- Get(ch, "now") returns the most recent timestamped item in a channel

To use the Get primitive, one would write:

<item1, ts1> = Get(ch1, <timestamp>); //timestamp can be a specific value "ts", "oldest", or "now"

a) [3 points] Please help Jiahao by filling in the timestamp fields in the following code snippet for accomplishing his desired outcome.

<item1, ts1> = Get(ch1, _____);
<item2, ts2> = Get(ch2, _____);
<item3, ts3> = Get(ch3, _____);

Ans:

<item1, ts1> = Get(ch1, "now"); // get the latest camera image
<item2, ts2> = Get(ch2, ts1); //get the audio sample closest to the timestamp associated with the camera image
<item3, ts3> = Get(ch3, ts1); //get the smoke sample closest to the timestamp associated with the camera image

b) [3 points] What timestamp values are returned by the above calls given the state of the channels?

ts1 = _____

ts2 = _____

ts3 = _____

Ans:

ts1 = 200

ts2 = 180

ts3 = 210

2. [2 points] (Answer True/False with justification – No points without justification)

In PTS, the same Get API can be used to retrieve both live and historical (archived) data.

Ans:

- [+1] True
- [+1] PTS provides a unified abstraction. Same Get works for both live and stored data

Failures & Recovery [25 points]

LRVM [9 points]

1. [2 points] In LRVM, there is the possibility of a "window of vulnerability". Concisely explain when and why this can occur in LRVM, and why this might be acceptable.

Ans:

[+1 point]

This can happen when "no-flush" mode is used in end_transaction, because LRVM doesn't block the process until the redo log is committed to disk. This leads to a time window between end_transaction and actual flush to the disk which is the window of vulnerability where data can be lost if there is either software failure or power failure.

AND

[+1 point]

Power failure is not very common. As long as the redo log is written to the disk in a bounded amount of time this may be an acceptable risk.

2. [3 points] Provide the three steps LRVM takes upon crash recovery.

Ans:

[+1 point per bullet]

- read the redo log from tail to head
- construct an in-memory tree of latest committed changes to the external data segments
- traverse the tree and apply the changes to the external data segment

3. [4 points] Suppose you decided to use LRVM in your application. The application starts a transaction and uses set_range to identify a block of memory. While in the middle of the operation (i.e., the application has modified some of the memory within that range, but not all), the system crashes. Before the crash, no end_transaction was called.

a) [2 points] Do you agree with the following statement: the persistent system state becomes inconsistent. Justify your answer.

Ans:

+1 for correct answer
+1 for correct justification

Disagree. External data segment is not updated at all through these writes. Since end_tx wasn't called the system behaves as if nothing happened.

b) [2 points] Compare the implication if the crash happens before end_transaction vs. after end_transaction

Ans:

+1 for each case

- In the first case, recover process does *nothing*.
- In the second case, the recovery process will update the external data segment to reflect the writes.

RioVista [12 points]

1. [2 points] A system with a battery-backed RIO file cache is running Satya's LRVM (not Vista). With a couple of succinct bullets,

explain whether this system would be more performant than running LRVM on a traditional file system without the RIO file cache [2 min]

Ans:

- [+2] This system is more performant while running LRVM. All the synchronous writes in RVM during failure free operation are absorbed by the RIO file cache. This includes the redo logs and the data segments that are mapped into memory

2. [10 points] A program using Vista and running on a system with the Rio file cache is trying to atomically update the variables foo and bar within a transaction, where both are within the specified data range. The initial value of each is set to TRUE. Within the transaction, foo is set to FALSE and later bar is set to FALSE. For each question, explain your choice using 1-2 succinct bullets. [5-7 min]

a) [2 points] The system crashes due to a kernel panic just after setting the value of foo and before setting the value of bar. What will be the persisted values of foo and bar before its crash recovery is started.

Ans:

- [+1] foo = FALSE, bar = TRUE.
- [+1] The values are persisted in the Rio file cache. Foo was set and bar was not. The undo log has yet to be applied.

b) [2 points] The system crashes just after setting the value of foo and before setting the value of bar. What will be the persisted values of foo and bar when the system is done with its crash recovery

Ans:

- [+1] TRUE for both.
- [+1] The state stored in the undo log had both set to TRUE. The variables have been restored to their initial state before the transaction started.

c) [2 points] The system crashes just after the end_xact() has finished. What will be the persisted values in foo and bar after the system recovers?

Ans:

- [+1] FALSE for both.
- [+1] In this case, the transaction was completed. The state has been persisted and the undo log has been removed. Vista will not apply any changes.

d) [2 points] The system crashes just after setting the value of foo. It then crashes again during the recovery process. Does the second crash affect the contents of the persisted data after recovery

Ans:

- [+1] No.
- [+1] Applying the UNDO log to the database during crash recovery is idempotent. So the crash during crash recovery only means that the UNDO log will be applied a second time to the database.

e) [2 points] The system crashes just after the end_xact() has finished. Will the application begin its execution after the end of the transaction

Ans:

- [+2] Rio Vista simply ensures that the data range is atomically persisted. It is up to the application developer to handle logic outside of state management, such as how and where an application is restarted.

Quicksilver [4 points]

1. [2 points] Comparing Quicksilver and LRVM, state which of the two provides a more comprehensive recovery management and why? (not more than two succinct bullet points

Ans:

- LRVM only provides recoverable virtual memory
- Quicksilver allows recovery of all systems resources that may have leaked (memory, network handles, file handles, orphan windows on display, etc.)

2. [2 points] Focusing on memory recovery, give two bullets to showcase why LRVM may potentially incur more unnecessary overhead for recovery management compared to Quicksilver.

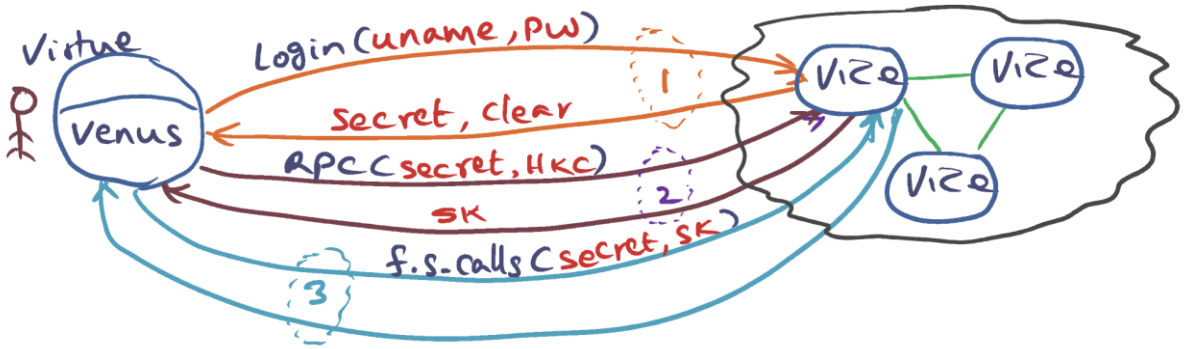
Ans:

- The undo record creation may be unnecessary (done by "Set range call") if no changes were made inside the transaction.
- Redo logs will be created when the transaction commits for the address range specified in the Set Range call even if no changes were made inside the transaction. The TM in Quicksilver only logs actual mods to persistent data

structures.

Security [6 points]

AFS [6 points]



1. [2 points] A "malicious" student writes a script to fetch 100,000 files from vice. Would vice allow this? If yes, state the implications of this. If not, explain why.

Ans: [+1] Yes, resource exhaustion is an issue in AFS. There is no confinement of resource usage for the users.
[+1] Large number of "genuine" requests can result in Denial-of-Service(DoS)

2. [2 points] In a public-key cryptographic system, the encryption key is published. Why is this not a security problem?

Ans: [+1] Public-key cryptography is based on one-way functions. The public key is good ONLY for encrypting. It cannot be used for decrypting.

[+1] For decrypting one needs a private key which is never exposed on the wire to anyone. It is directly given (out of band) to the specific individuals/systems who have the authority to decrypt the message

3. [2 points] Your friend argues that the bind protocol is unnecessary to execute within each RPC session once a legitimate user has successfully logged into AFS. Instead, she proposes that Virtue can generate a new session key and communicate that to Vice for each RPC session. Virtue can then use that session key for the duration of the RPC session. Is there a flaw in her proposal? Answer Yes/No with justification.

Ans: [+2 points] No. In principle this can also work to avoid the over-exposure of the handshake key established during the login session

Kishore's note: If the student answers Yes, then we have to see the justification. An acceptable justification is that Vice is secure while Virtue is not.

HIDDEN - Potpourri [14 points]

[Multiple choice for hidden Potpourri]

1. [2 points] Through the special nature of the Login process interaction between Virtue and Vice, an ephemeral identity and a handshake key to use during that Login session is given by Vice to the Virtue workstation for a legitimate user of AFS.

Given this, why can't Virtue simply use this ephemeral identity and encrypting all the subsequent file system calls that the user makes to the file server using the handshake key?

(Multiple choice -- choose 1 by specifying its number)

i) To ward off replay attacks

ii) Since the handshake key is for one-time use by Virtue

iii) To reduce over-exposure of the handshake key within the login session

iv) Since the handshake key is valid only for a limited time

v) Since the handshake key is valid only for a limited number of calls to the file server

2. [6 Points] You are working on a large financial data system. This system contains a complete view of trades ingested from various exchanges. The primary feature is a read heavy, highly detailed dashboard that shows all open positions, and offers various search filters for analyzing relevant trades. The system's data is updated in batches via intraday SFTP files delivered every 5-30 minutes, depending on the exchange.

Assumptions and Considerations:

- Each dashboard request may need to consult the full corpus of data
- A partial answer may be considered unsafe
- The system uses full replication on each node

a) [1 point] What can you sacrifice if there is a spike (i.e., can you sacrifice both yield and harvest, or only yield, or only harvest) and how? Justify your answer in 1-2 sentences.

Ans: [+1] You can only sacrifice yield. Since partial answers may be unsafe, harvest must be protected at the cost of yield.

b) [2 points] Suppose instead of waiting for the intraday files to see new trades in batches, the system starts to receive updates about all trades in real time. The dashboard should still display all the same information but now have access to real time trade data.

In 1-2 bullets, explain why full replication becomes a bottleneck for this shift in workload. Be specific about what must happen on each write operation.

Ans:

[+1] Full Replication is fine under infrequent batch writes but becomes difficult to scale under write heavy workloads.

[+1] Since the data is replicated on each node, writes need to propagate to each node of the system, increasing network traffic and making it harder to scale.

c) [3 points] Propose a redesign that improves write scalability while maintaining fault tolerance? Your answer should include 3 bullet points that clearly specify what replication strategy you will use, how data is distributed, and how fault tolerance is preserved.

Ans:

[+1] Replication: Maintain some N >= 2 replicas of each partition for fault tolerance.

[+1] Data Distribution: Partition data across nodes for better write performance. Ex partition by exchange or account ID.

[+1] Fault Tolerance: Each partition is replicated, so no single node failure causes data loss

3. [2 points] QuickSilver's Log Manager (LM) provides a common log, but explicitly requires servers to independently scan the log and drive their own recovery.

a) [1 point] State exactly **ONE advantage** of this decoupled approach in exactly one sentence.

Ans:

- Any valid advantage and valid disadvantage gets full credit.
- [+1] **Advantage:** Each server can tailor their own recovery technique rather than one-size-fits-all

b) [1 point] State exactly **ONE mitigation** technique QuickSilver uses to reduce the read-performance penalty of servers having to scan the log themselves in exactly one sentence.

Ans: [+1] **Mitigation:** The LM exposes an index into the log for each server's log records.

(Note: Acceptable alternate mitigations based on the paper include: "The LM uses scan directional information to prefetch data blocks" OR "The LM maintains backpointers on log records to minimize the cost of backward scans.")

4. [2 points] (Answer True/False with justification – No points without justification)

PTS channels enforce FIFO ordering of items, meaning items are always retrieved in the order they were inserted.

Ans:

- [+1] False
- [+1] PTS is time-based, not FIFO-based. Retrieval depends on timestamps, not insertion order

5. [2 points] (Answer True/False with justification – No points without justification)

In PTS, Put calls from multiple threads into the same channel requires the threads to acquire a mutual exclusion lock for the channel access

Ans:

- False
- Items in the channel are immutable. The put calls from the threads are independent events and result in each item being placed in a time-ordered manner into the same channel.